

YIDAN ZHU

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EDUCATION

Southern University of Science and Technology (SUSTech) **Expected Jun. 2028**
B.S. in Optoelectronic Information Science and Engineering
GPA: 3.94/4.00 | **Major Rank:** 2/42 | Wolong Innovation Honors Program

RESEARCH INTERESTS

Photonic crystals, optical computing, and optical communications

RESEARCH EXPERIENCE

Undergraduate Researcher | Anisotropy Control of Dirac Cones **Jul. 2026 - Present**
Prof. Jingwen Ma's Group, SUSTech

- Investigate how dielectric anisotropy regulates Dirac-cone formation, migration, tilting, gap opening, and modal coupling in 3D photonic-crystal slabs.
- Built FDTD models and Python/LSF automated parameter sweeps for anisotropic dielectric tensors; extracted band structures, located Dirac points, quantified cone tilt, and analyzed 2D momentum-space dispersions and modal fields.

Undergraduate Researcher | Liquid-Crystal Photoalignment and Optical Characterization **Jul. 2025 - Mar. 2026**
Prof. Dan Luo's Group, SUSTech

- Prepared and characterized SD1-based liquid-crystal samples using spin coating, patterned UV exposure, polarized optical microscopy.
- Initially explored self-organized helical structures, then systematically varied SD1 concentration, DMD exposure time, and liquid-crystal layer thickness to analyze angle-dependent color, intensity, and transmittance for multidimensional encrypted displays.

SELECTED PROJECT

Photonic-Crystal Visuotactile Sensor | SUSTech **Jul. 2025 - Aug. 2025**
Prof. Dan Luo's Group

- Participated in the full fabrication process, including layered spin coating on glass, deposition of white tracking markers, encapsulation, and integration with a handheld camera-and-illumination setup.

PUBLICATION

Liu, M., Wu, Z., Zheng, J., **Zhu, Y.**, Liu, Y., Ma, L., Hu, W., Lu, Y.-Q., & Luo, D. (2026). From 0D to 3D Liquid Crystal Elastomers Architectures: Fabrications and Applications. *Responsive Materials*, 4(2), e70042. <https://doi.org/10.1002/rpm2.70042>

HONORS AND AWARDS

National Second Prize (Team) — China Youth Science and Technology Innovation Competition 2025
An Interactive Domain-Specific Compiler for Signal-Processing Algorithm Mapping

Third Prize (Team) — SUSTech International Innovation and Entrepreneurship Competition Oct. 2025
Ultrathin, High-Precision Photonic-Crystal Visuotactile Sensor

Outstanding Student Honor 2025

Outstanding Student Scholarship, Second Prize 2025

SKILLS

Simulation: Ansys Lumerical FDTD | **Programming & scripting:** Python, Lumerical Script (LSF), MATLAB, C
Experimental: Spin coating, photoalignment, UV exposure, DMD operation, polarized optical microscopy, transmission spectroscopy, material dispensing, encapsulation
Language: TOEFL iBT 100/120 (CEFR C1), Sep. 2026